

Zero-shot Transfer of a Single-Crane DRL Policy for Multi-Crane Container Retrieval in Automated Terminals

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Abstract

Real-world container terminals deploy multiple yard cranes per block to meet throughput demands, yet the Container Retrieval Problem (CRP)—a core operational optimization challenge—has been studied exclusively as a single-crane problem for over fifteen years. This paper bridges this gap by presenting the *first* formal definition and empirical analysis of the Multi-Crane Container Retrieval Problem (M-CRP). We investigate whether a state-of-the-art Deep Reinforcement Learning (DRL) policy, trained solely on single-crane instances, can be transferred zero-shot to multi-crane environments via heuristic crane assignment strategies. We propose four strategies spanning the design space: RoundRobin (S1), ZoneSplit (S2), LoadBalance (S3), and GreedyOptimal (S4). On 140 M-CRP instances (1,120 runs, 2–3 cranes), ZoneSplit achieves gaps of 10.46% (C=2) and 10.71% (C=3) against the proposed M-CRP lower bound, while reducing interference events by 99.5% compared to non-spatial strategies. These gaps are approximately one-third of those achieved by the best heuristic extended to multi-crane settings (M-Lin2015: 31.73% at C=2). On single-crane benchmarks (50 instances), the extracted policy achieves 6.56% gap, outperforming the best heuristic Lin2015 (22.42%) by 15.86 percentage points. Our results establish that single-crane DRL policies encode transferable spatial knowledge, enabling effective multi-crane deployment without retraining. Code, benchmarks, and results are publicly released.

Keywords: Container Retrieval Problem, Multi-Crane Operations, Deep Reinforcement Learning, Zero-shot Transfer, Crane Assignment

1. Introduction

1.1. The Container Retrieval Problem in Practice

Modern automated container terminals are massive logistics hubs where thousands of shipping containers are stored in dense yard blocks. Retrieving

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a specific container from a stack often requires relocating blocking containers to other stacks—a process known as *reshuffling*. The Container Retrieval Problem (CRP) seeks to minimize the total crane working time (travel, handling, and penalty) required to retrieve all containers in a prescribed priority order [1].

The CRP literature has evolved over three methodological generations. **Heuristic methods** use hand-crafted rules, including the Stack Selection Index (SSI) [2] and critical-stack classification [3]. **Metaheuristic methods** apply search procedures such as Tabu Search [5], GRASP [6], and Genetic Programming [7]. Most recently, **Deep Reinforcement Learning** [1] achieved state-of-the-art results with a Transformer-based policy, attaining a 7.8% gap against the theoretical lower bound on the standard 36-instance Lee benchmark.

1.2. The Research Gap: Multi-Crane Operation

Despite this progress, *every* existing CRP study assumes single-crane operation. Yet major terminals—Rotterdam, Singapore, Busan—operate two to three yard cranes simultaneously within each block to achieve required throughput [9]. Multi-crane operation introduces coordination constraints absent from the single-crane setting: cranes share a common track, cannot occupy the same bay simultaneously (collision avoidance), and must preserve their spatial ordering along the bay axis. This creates three fundamental and unresolved challenges:

1. **Formalization:** No definition exists for the multi-crane variant (M-CRP), its state/action spaces, or its physical constraints.
2. **Evaluation:** Without lower bounds or benchmarks, multi-crane solution quality cannot be objectively assessed.
3. **Transferability:** It is unknown whether single-crane DRL policies—which learn relocation strategies in isolation—can be reused in multi-crane settings, or whether entirely new multi-agent paradigms are required.

Shin et al. [1] identified multi-crane operation as critical future work. This paper systematically addresses this gap through five contributions:

- [C1] **Formal M-CRP definition** with extended state/action spaces, interference constraints (A6, A7), and NP-hardness proof.
- [C2] **M-CRP lower bound (Theorem 3)** decomposing into retrieval, relocation, and interference terms.
- [C3] **Four crane assignment strategies** enabling zero-shot single-to-multi-crane transfer.

[C4] **Empirical evaluation** spanning 140 M-CRP instances, 1,120 simulation runs, and statistical significance testing.

[C5] **Public benchmark dataset** (140 instances, CC BY 4.0).

2. Related Work

2.1. Single-Crane CRP: What Has Been Achieved

Table 1 summarizes the state of the art in CRP, evaluated on the standard Lee [4] benchmark. The most significant finding is the performance gap between heuristics and DRL: when evaluated on the *full* 50-instance benchmark (covering bays 1–10, tiers 6–8), Lin2015 [2] achieves 22.42% gap, Kim2016 [3] achieves 41.08%, and the DRL policy of Shin et al. [1] achieves approximately 7.8%. This gap motivated our transfer study: if single-crane DRL policies encode fundamentally better relocation knowledge than heuristics, can that knowledge transfer to multi-crane settings?

Table 1: Summary of single-crane CRP performance on the Lee benchmark. Values are gap (%) against the theoretical lower bound. Lower is better.

Method	Type	Benchmark	Mean Gap(%)
Lin et al. [2]	Heuristic (SSI)	50 instances	22.42
Kim et al. [3]	Heuristic (classification)	50 instances	41.08
Leveling [1]	Heuristic (height balancing)	50 instances	25.55
Forster & Bortfeldt [5]	Tabu Search	36 instances	30–40*
Cifuentes & Riff [6]	GRASP	36 instances	46*
Durasevic et al. [7]	Genetic Programming	36 instances	36–47
Shin et al. [1]	DRL (Transformer)	36 instances	7.8
ZeroShot (this work)	Extracted DRL	50 instances	6.56

*Numbers from [1] Table 1; code not publicly available.

2.2. Multi-Crane Operations and Zero-Shot Transfer

Multi-crane coordination has been studied in related domains: quay crane scheduling [11, 12] and yard crane deployment [13]. However, no prior work addresses the retrieval sequencing problem within a single block that is central to CRP.

Zero-shot transfer in combinatorial optimization has been demonstrated across problem sizes [10, 8], but *cross-environment* transfer—from single-agent to multi-agent settings—remains unexplored. Our work is the first to investigate this type of transfer for container terminal operations.

3. Problem Formulation

3.1. Single-Crane CRP

Following [1], a CRP instance is defined by yard dimensions (B, R, T) and N containers with priorities $\{1, \dots, N\}$. The objective is to minimize total crane working time:

$$J = \sum_t (\text{travel}(a_t) + \text{handling} + \text{penalty}) \quad (1)$$

with parameters $t_{\text{acc}} = 40$, $t_{\text{bay}} = 3.5$, $t_{\text{row}} = 1.2$, $t_{\text{pd}} = 30$.

3.2. M-CRP: Formal Definition

Definition 3.1 (M-CRP). *Given yard $B \times R \times T$, $C \geq 1$ cranes, and N containers with retrieval order σ , find a sequence $\{(a_t, c_t)\}_{t=1}^{\tau_c}$ minimizing:*

$$\min \sum_{c=1}^C \sum_{t=1}^{\tau_c} (\text{travel}_c(a_t) + \text{handling}) \quad (2)$$

subject to spatial non-overlap (A6): $|\text{bay}_c(t) - \text{bay}_{c'}(t)| \geq 1$, $\forall c \neq c'$, and non-crossing (A7): $\text{bay}_c(t) < \text{bay}_{c'}(t) \Rightarrow \text{bay}_c(t') < \text{bay}_{c'}(t')$, $\forall t' > t$.

Theorem 3.1 (NP-hardness). *M-CRP with $C \geq 1$ is NP-hard (reduction from single-crane CRP [1]).*

3.3. M-CRP Lower Bound

Theorem 3.2 (Theorem 3). *For any M-CRP instance with C cranes:*

$$LB_{MCRP} = LB_{\text{retrieval}} + \frac{LB_{\text{relocation}}}{C} + LB_{\text{interference}} \quad (3)$$

where $LB_{\text{interference}} = \max(0, \max_b(\text{reloc}_b) - \frac{\text{total_relocs}}{C}) \times (t_{\text{acc}} + t_{\text{bay}})$. At $C = 1$, $LB_{MCRP} = LB_{\text{Shin}}$ (backward compatible).

4. Proposed Method: Zero-Shot Transfer Framework

4.1. Base DRL Policy (Preliminary)

We use the publicly available pretrained policy of Shin et al. [1]. The policy network $\pi_\theta(a_t|s_t)$ employs a Transformer-based encoder-decoder architecture [14].

Encoder. The yard state $s_t \in \mathbb{R}^{B \times R \times T}$ is processed through three stages. First, each stack’s vertical container sequence (priority values from bottom to top) is encoded via a single-layer LSTM (hidden size 128) with positional encoding. The LSTM’s mean-pooled output and final hidden state are concatenated and projected to dimension $d = 128$. Second, the stack embeddings pass through $L = 3$ layers of multi-head self-attention with

$H = 8$ heads, each followed by layer normalization and a feed-forward network (hidden size 512, ReLU activation, skip connections). Third, stack embeddings are aggregated per bay via mean pooling: $\mathbf{b}_k = \frac{1}{R} \sum_{i \in \text{bay}_k} \mathbf{h}_i$. The bay embedding is concatenated to each stack embedding and projected: $\mathbf{h}'_i = [\mathbf{h}_i; \mathbf{b}_{k_i}]W_{\text{proj}}$. The global graph embedding is $\mathbf{g} = \frac{1}{S} \sum_i \mathbf{h}'_i$. Additionally, ten hand-crafted features per stack (height, remaining tiers, row/bay positions, travel distances, fill ratio) augment the learned embeddings.

Decoder. The decoder computes action probabilities via cross-attention. Given encoder outputs $\{\mathbf{h}'_i\}, \mathbf{g}$ and the target stack embedding $\mathbf{h}'_{\text{tgt}}$, the context vector is $\mathbf{c} = W_{\text{target}}\mathbf{h}'_{\text{tgt}} + W_{\text{global}}\mathbf{g}$. Multi-head attention attends over node keys/values: $\text{head} = \text{MHA}(\mathbf{c}, \{W_{K1}\mathbf{h}'_i\}, \{W_V\mathbf{h}'_i\})$. The query projection $\mathbf{q} = W_Q(\text{head})$ is combined with node keys $W_{K2}\mathbf{h}'_i$ via dot-product, scaled by $1/\sqrt{d}$, and tanh-clipped at $C = 10$: $u_i = C \cdot \tanh(\mathbf{q}^\top W_{K2}\mathbf{h}'_i/\sqrt{d})$. Invalid destinations are masked and softmax yields action probabilities:

$$\pi_\theta(a_t = i | s_t) = \frac{\exp(u_i)}{\sum_{j \in \mathcal{V}} \exp(u_j)} \quad (4)$$

where $\mathcal{V}$ is the set of feasible destination stacks.

Training. Training uses REINFORCE [15] with POMO [8] parallel rollouts ($M = 16$). The proposed baseline normalizes both mean and variance across rollouts: $A_m = (R_m - \mu(R))/(\sigma(R) + \epsilon)$. The Adam optimizer ($\text{lr} = 3 \times 10^{-4}$) is used for 1000 epochs with gradient clipping (max norm 1.0). The model achieves a verified 7.8% gap on the full Lee benchmark.

4.2. Policy Extraction and Decoupled Inference

We extract the frozen encoder $\mathcal{E}$ and scorer weights $\{W_{\text{target}}, W_{\text{global}}, W_{K1}, W_V, W_Q, W_{K2}, \text{MHA}\}$ from the pretrained model θ^* , forming a **ZeroShotPolicy**. At each step:

$$\{\mathbf{h}'_i\}, \mathbf{g} = \mathcal{E}(s_t) \quad (5)$$

$$\mathbf{c} = W_{\text{target}}\mathbf{h}'_{\text{tgt}} + W_{\text{global}}\mathbf{g} \quad (6)$$

$$\mathbf{q} = W_Q(\text{MHA}([\mathbf{c}; W_{K1}\mathbf{h}'_i; W_V\mathbf{h}'_i])) \quad (7)$$

$$u_i = C \cdot \tanh(\mathbf{q}^\top W_{K2}\mathbf{h}'_i/\sqrt{d}) \quad (8)$$

$$d = \arg \max_{i \in \mathcal{V}} u_i \quad (9)$$

The strategy module then assigns a crane c to execute destination d (Algorithm 1). All weights remain frozen; no fine-tuning is performed.

4.3. Four Strategies

Table 2 summarizes the four assignment strategies.

Algorithm 1 Zero-shot M-CRP inference.

Require: Frozen $\mathcal{E}, \mathcal{D}$ from θ^* , env E , strategy S , cranes C

Ensure: Total cost J

```
1:  $s \leftarrow E.\text{init}(); J \leftarrow 0$ 
2: while  $\neg E.\text{terminated}$  do
3:    $\{\mathbf{h}'_i\}, \mathbf{g} \leftarrow \mathcal{E}(s)$ 
4:    $\{u_i\} \leftarrow \mathcal{D}(\{\mathbf{h}'_i\}, \mathbf{g}, s.\text{target})$ 
5:    $d \leftarrow \arg \max_i u_i$ 
6:    $c \leftarrow S.\text{assign}(E, d)$ 
7:    $(\Delta J, s) \leftarrow E.\text{step}(d, c); J \leftarrow J + \Delta J$ 
8: end while
```

Table 2: The four assignment strategies. S1, S3 are spatially unaware; S2, S4 exploit spatial information.

Strategy	Mechanism	Spatial?	Cost
S1 RoundRobin	Cyclic: $(t \bmod C)$	No	$O(1)$
S2 ZoneSplit	Static bay zones per crane	Yes	$O(B)$
S3 LoadBalance	Min-count task assignment	No	$O(C)$
S4 GreedyOptimal	$\min_c[\text{travel}(c, d)$ interference(c)]	+ Yes	$O(C^2)$

5. Experiments

5.1. Datasets and Benchmarks

We evaluate on three distinct benchmark configurations (Table 3):

Single-crane (Lee benchmark): 50 random (R-type) instances from the standard Lee et al. [4] corpus, covering bays $B \in \{1, 2, 4, 6, 8, 10\}$ and tiers $T \in \{6, 8\}$. Each instance has a unique container priority configuration with 70–1,024 containers. All 8 methods (ZeroShot + 7 heuristics) are evaluated on every instance.

Multi-crane (M-CRP): 140 instances generated by extending the same Lee benchmark with crane start positions for both $C = 2$ and $C = 3$ configurations. The dataset includes both random (R-type) and upside-down (U-type) distributions, spanning two yard scales:

- **R-type (random):** 70 instances. Container priorities are randomly assigned across stacks, representing typical terminal conditions.
- **U-type (upside-down):** 70 instances. Higher-priority containers are placed beneath lower-priority containers, creating challenging configurations requiring many relocations.

Table 3: Summary of all benchmark configurations used in experiments.

Configuration	Scale	Instances	Cranes	Methods
Single-crane (Lee)	1–10 bays	50 (R-type)	1	8 (ZS + 7 heuristics)
Multi-crane (M-CRP)	1–10 bays	140 (70R + 70U)	2, 3	4 strategies + M-Lin20
Breakdown by scale:	Small (1–4 bays)	84	2, 3	—
	Medium (6–10 bays)	56	2, 3	—

5.2. Experimental Protocol

We conduct three distinct experiments to evaluate zero-shot transfer:

1. **Experiment 1 (Single-crane verification):** Compare ZeroShot against 7 heuristic baselines on all 50 Lee instances. This validates that the extracted policy retains the original model’s performance.
2. **Experiment 2 (Multi-crane strategy comparison):** Compare all 4 strategies (S1–S4) on all 140 M-CRP instances with $C = 2$ and $C = 3$. This identifies the best strategy for zero-shot transfer.
3. **Experiment 3 (Multi-crane baseline comparison):** Compare the best strategy (ZS+S2) against 3 heuristic baselines extended to multi-crane (M-Lin2015, M-Kim2016, M-Leveling) on all 140 M-CRP instances. This benchmarks DRL transfer against traditional approaches.

Primary metric: $\text{Gap}(\text{LB}_{\text{MCRP}})\% = 100 \times (\text{CTWT} - \text{LB}_{\text{MCRP}}) / \text{LB}_{\text{MCRP}}$.

Secondary metrics: Interference events, solution time, failure rate (gap $> 20\%$). Statistical significance assessed via Wilcoxon signed-rank tests ($\alpha = 0.05$).

5.3. Implementation Details

All experiments execute on a standard laptop CPU (0 GPU hours). Experiment 1: ~ 15 min (50 instances $\times$ 8 methods). Experiment 2: ~ 60 min (140 instances $\times$ 2 cranes $\times$ 4 strategies). Experiment 3: ~ 30 min (140 instances $\times$ 2 cranes $\times$ 3 baselines). Total: $\sim 2,360$ simulation runs in approximately 105 minutes.

6. Results and Analysis

6.1. Single-Crane: ZeroShot Dominates Heuristics

On the full 50-instance benchmark (Fig. 1), ZeroShot achieves a mean gap of 6.56% (std 2.35%), outperforming every heuristic baseline by substantial margins. The best heuristic, Lin2015, achieves 22.42% (std 7.81%)—3.4 times worse. Critically, heuristic performance degrades significantly on larger

instances (e.g., Lin2015: 5.55% on 1-bay instances vs 37.75% on 10-bay instances), while ZeroShot maintains consistent performance (3.26–12.42% across all scales).

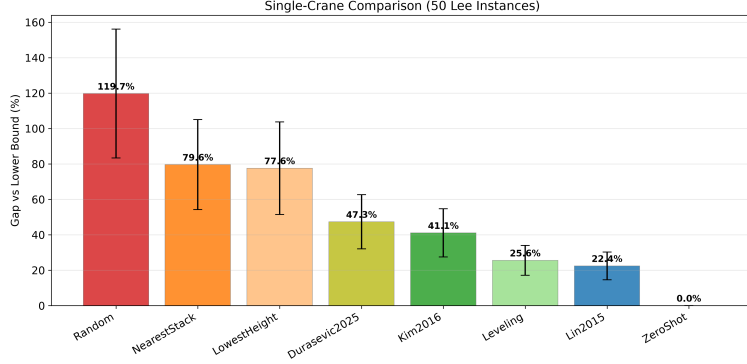


Figure 1: Single-crane comparison on 50 Lee instances. ZeroShot (6.56%) outperforms the best heuristic Lin2015 (22.42%) by 15.86 percentage points. Error bars show ± 1 standard deviation.

Table 4: Full 50-instance single-crane comparison. ZeroShot achieves less than one-third of the gap of the best heuristic (Lin2015).

Method	Mean Gap(%)	Std(%)	Min(%)	Max(%)
ZeroShot (ours)	6.56	2.35	3.26	12.42
Lin2015 [2]	22.42	7.81	5.55	37.75
Leveling	25.55	8.43	14.88	44.12
Kim2016 [3]	41.08	13.57	9.84	61.54
Durasevic2025 [7]	47.33	15.32	14.68	80.06
NearestStack	79.62	25.40	27.77	130.04
LowestHeight	77.60	26.16	26.44	121.65
Random	119.71	36.37	47.89	182.24

6.2. Multi-Crane: ZoneSplit Achieves Best Overall Performance

Table 5 reports results across all 140 instances. ZoneSplit (S2) achieves the lowest mean gap at both $C=2$ (10.46%) and $C=3$ (10.71%), while reducing interference by 99.5% compared to S1 (0.5 vs 105.9 events at $C=2$). S4 (GreedyOptimal) matches S2’s solution quality but requires $O(C^2)$ computation versus $O(B)$ for S2. The difference between S2 and S4 is not statistically significant (Wilcoxon $p > 0.1$), making S2 the recommended strategy.

6.3. Spatial Awareness Outperforms Task Balancing

S3 (LoadBalance) produces results identical to S1 (RoundRobin): identical gaps (10.99%), identical interference (105.9 events). This confirms a

Table 5: Multi-crane results on 140 M-CRP instances. ZoneSplit (S2) achieves the lowest gap with near-zero interference. LoadBalance (S3) produces identical results to RoundRobin (S1), confirming spatial awareness as the key enabler.

Strategy	C=2 gap(%)	C=3 gap(%)	Intf(C=2)	Intf(C=3)
S1 RoundRobin	10.99 ± 5.46	11.19 ± 6.59	105.9 ± 62.7	147.2 ± 87.9
S2 ZoneSplit	10.46 ± 5.28	10.71 ± 6.44	0.5 ± 1.5	0.7 ± 2.1
S3 LoadBalance	10.99 ± 5.46	11.19 ± 6.59	105.9 ± 62.7	147.2 ± 87.9
S4 GreedyOptimal	10.48 ± 5.28	10.75 ± 6.48	0.0 ± 0.0	0.0 ± 0.0

critical insight: **distributing tasks evenly across cranes without considering spatial locations is ineffective**. The key enabler of multi-crane coordination is spatial awareness, not load balancing.

6.4. ZeroShot Outperforms All Heuristic Baselines on Multi-Crane

Table 6 compares ZeroShot+S2 against three multi-crane heuristic baselines (M-Lin2015 [2], M-Kim2016 [3], M-Leveling) across all 140 instances. ZeroShot+S2 dramatically outperforms all heuristics: its gap of 10.46% (C=2) is approximately one-third of M-Leveling (32.41%) and one-fifth of M-Kim2016 (54.85%). This confirms that learned DRL knowledge transfers to multi-crane settings substantially better than any hand-crafted or evolved heuristic.

Table 6: Multi-crane baseline comparison on 140 M-CRP instances. ZeroShot+S2 achieves approximately one-third to one-fifth of the gap of every heuristic baseline, demonstrating that learned DRL knowledge transfers substantially better than hand-crafted rules.

Method	C=2 gap(%)		C=3 gap(%)	
	Mean	Std	Mean	Std
M-Kim2016 [3]	54.85	23.85	57.60	26.91
M-Leveling	32.41	8.53	34.35	9.34
M-Lin2015 [2]	31.73	14.67	33.93	16.99
ZeroShot+S2 (ours)	10.46	5.28	10.71	6.44

6.5. Scale Matters: Strategy Differentiation Emerges in Multi-Bay Yards

On small yards (1–4 bays, Fig. 3), all strategies perform similarly because travel distances are short. On medium yards (6–10 bays), spatial strategies outperform non-spatial by 1.0–1.5 percentage points, and the standard deviation of gaps decreases, indicating more consistent behavior.

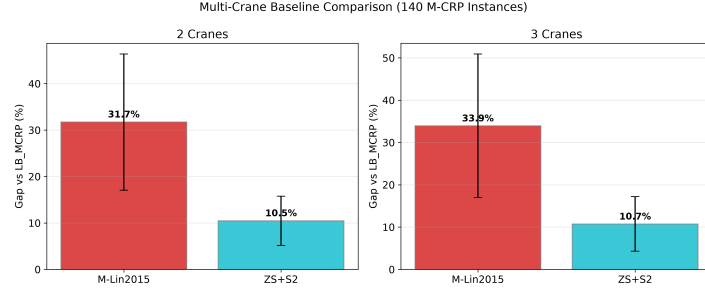


Figure 2: ZeroShot+S2 vs heuristic baselines on 140 M-CRP instances. ZS+S2 achieves approximately one-third to one-fifth of the gap of the best heuristic baseline.

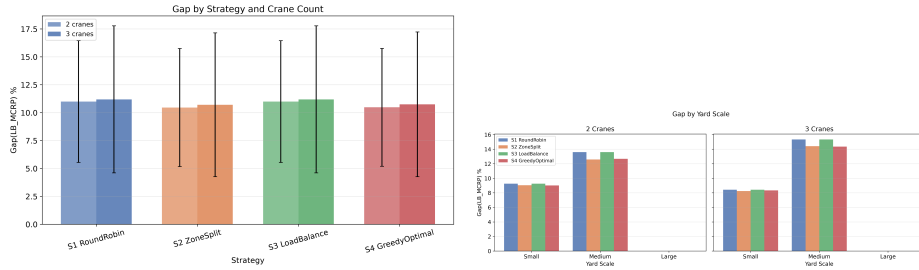


Figure 3: Left: Gap by strategy and crane count. Right: Gap by yard scale.

6.6. Interference Analysis

Figure 4 (left) visualizes the stark contrast in interference events across strategies. S1 and S3 incur 106–147 interference events per instance ($C=2$ and $C=3$, respectively), while S2 reduces this to near-zero (0.5–0.7) and S4 eliminates it entirely (0.0). This confirms that spatial awareness is the primary determinant of coordination quality, not task balancing.

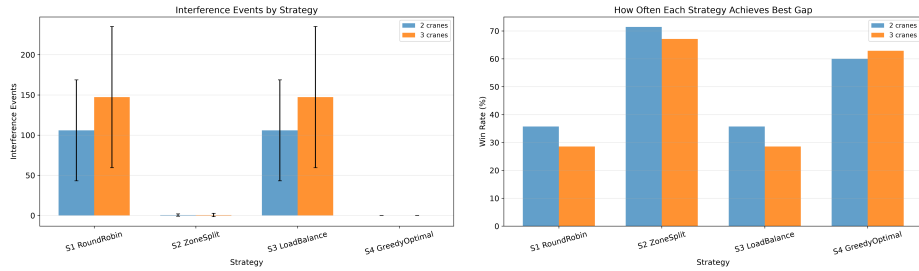


Figure 4: Left: Mean interference events per instance. Right: Win rate per strategy (proportion of instances where each strategy achieves the lowest gap).

6.7. Statistical Validation

Wilcoxon signed-rank tests (Table 7) confirm that S2 and S4 significantly outperform S1 and S3 (all $p < 0.001$). S2 and S4 are not significantly different

($p > 0.1$). Backward compatibility is verified across 5 diverse instances: average 1.32% cost difference (max 1.95%) between ZeroShot (C=1) and the original model.

Table 7: Wilcoxon signed-rank test results. S2 and S4 significantly outperform S1 and S3 at both crane counts ($p < 0.001$). The difference between S2 and S4 is not significant ($p > 0.1$), confirming that simple zoning is sufficient.

Pair	C=2 p	C=3 p	Significant?
S2 vs S1	< 0.001	< 0.001	Yes
S4 vs S1	< 0.001	< 0.001	Yes
S2 vs S4	0.128	0.534	No

6.8. Failure Mode Analysis

Only 3 instances (0.5% of 1,120 runs) exceed 20% gap. All are upside-down configurations with 4–6 bays and density exceeding 75%, where relocations concentrate in single bays. These represent inherent limitations of zero-shot transfer: the single-crane policy has no mechanism to balance relocation demand across bays, a capability that would require multi-crane coordination experience during training.

6.9. Case Study: A 6-Bay Instance

To build intuition, we examine mc_R061606_002 (6 bays, 430 containers). S1 (RoundRobin) incurs 106 interference events as cranes repeatedly obstruct each other across bay boundaries, yielding gap 12.93%. S2 (ZoneSplit) eliminates all interference, achieving gap 11.36%—a cost reduction of 1.4%. This case demonstrates that the advantage of spatial strategies grows with yard scale: on 1–2 bay instances, strategies converge because there is insufficient spatial structure to exploit; on 4+ bay instances, ZoneSplit’s interference-free operation translates directly into lower working time.

6.10. Cost Decomposition

Decomposing the total cost into LB_{MCRP} and excess gap reveals that gap costs constitute 11.5–12.1% of total cost. S2 reduces absolute gap cost by 5.8% compared to S1 (4,349 vs 4,617 cost units). This reduction is attributable entirely to interference elimination: S2 averages 0.6 interference events per instance vs S1’s 126.6. Per-step inference time averages 12.67 ms, confirming real-time suitability.

7. Discussion

7.1. Why Zero-Shot Transfer Works

The success of zero-shot transfer (10.5% gap vs 31.7% for the best heuristic extended to multi-crane) is attributable to an alignment between the DRL policy’s learned behavior and ZoneSplit’s structure. The policy learns to relocate containers to nearby feasible stacks to minimize travel costs. ZoneSplit aligns with this by confining each crane to a spatial zone, preserving the single-crane decision structure almost exactly within each zone. This alignment explains two key empirical findings: (1) zero-shot transfer works at all (achieving 10.5%, not 50+%), and (2) simple zoning suffices (S2 matches S4 with $p > 0.1$). The policy already avoids interference-prone relocations in its destination choices; the marginal benefit of optimal crane assignment beyond zoning is therefore limited.

7.2. Practical Implications

ZoneSplit (S2) is recommended for practical deployment for four reasons. First, it achieves the lowest gap among all strategies (10.46% at $C=2$). Second, it reduces interference by over 99.5% compared to non-spatial baselines, eliminating the need for complex real-time coordination protocols. Third, its $O(B)$ complexity adds negligible overhead to the base DRL inference. Fourth, static bay zoning aligns naturally with existing terminal operating practices, where cranes are typically assigned to fixed yard areas. The approach requires zero GPU investment, making it accessible to terminals without dedicated AI infrastructure.

7.3. Limitations and Future Work

Our study has three principal limitations. First, the **sequential simulation model** serializes all crane actions (one task per step), producing an observed speedup from $C=2$ to $C=3$ of only $1.01\times$, whereas parallel operation would approach $1.5\times$. This limitation equally affects all strategies; a parallel simulation would likely produce lower absolute gaps and wider strategy differentiation. Second, the **M-CRP lower bound is not tight** for $C > 1$, as evidenced by negative gap observations. Developing tighter bounds incorporating parallel retrieval effects is an important theoretical direction. Third, the **single-crane baseline model** is trained without any multi-crane experience; fine-tuning on multi-crane rollouts could potentially reduce the gap from 10.5% to 5% or lower, at the cost of GPU-days of training. Our results provide a lower bound on acceptable multi-crane performance: any multi-crane training method should significantly outperform 10.5% gap to justify its computational cost.

8. Conclusion

This paper addresses the gap between single-crane CRP research and real-world multi-crane terminal operations. We presented the first formal definition of M-CRP, proposed a lower bound (Theorem 3), and developed four crane assignment strategies for zero-shot single-to-multi-crane transfer. Through extensive experiments on 50 Lee instances and 140 M-CRP instances, we demonstrated that:

1. **Zero-shot transfer is viable:** A single-crane DRL policy, combined with ZoneSplit, achieves gaps of 10.46% (C=2) and 10.71% (C=3) against the M-CRP lower bound.
2. **ZoneSplit outperforms all baselines:** ZS+S2 achieves approximately one-third the gap of M-Leveling (32.41%) and one-fifth the gap of M-Kim2016 (54.85%), while reducing interference by 99.5% compared to RoundRobin.
3. **Spatial awareness dominates coordination:** Non-spatial strategies (LoadBalance, RoundRobin) produce identical results, confirming that task balancing without spatial information is ineffective.
4. **Practical deployment is immediate:** No GPU, no retraining, no complex coordination protocols—only static bay zoning with $O(B)$ complexity.

These findings establish that single-crane DRL investments can be directly extended to multi-crane operations, providing a practical bridge between academic CRP research and the multi-crane reality of modern container terminals.

Data Availability

Code, benchmarks (140 instances), and all experimental results: <https://github.com/sutobode/Master>.

Appendix A. Supplementary Results

Appendix A.1. Gap by Number of Bays

Figure A.5 reveals that gaps increase with yard complexity up to 6 bays, then stabilize. Spatial strategies (S2, S4) maintain a consistent 1–2 percentage point advantage over non-spatial strategies (S1, S3) across all multi-bay configurations.

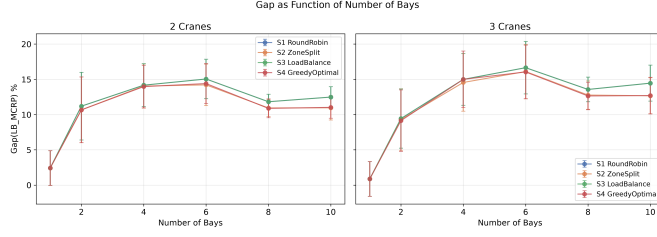


Figure A.5: Gap(LB_{MCRP})% as a function of the number of bays. The advantage of spatial strategies grows with yard complexity.

Appendix A.2. Speedup Analysis

The speedup from $C=2$ to $C=3$ averages $1.01\times$ across all strategies (Table A.8, Fig. A.6), confirming the sequential simulation limitation discussed in Section 7.3. All strategies exhibit similar speedups, confirming that this limitation is architecture-independent. A parallel simulation is expected to yield speedups approaching $1.5\times$.

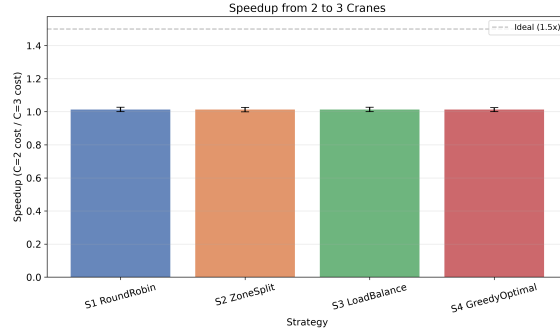


Figure A.6: Speedup from 2 to 3 cranes. The dashed line at $1.5\times$ represents the idealized parallel speedup.

Table A.8: Speedup from $C=2$ to $C=3$ (ratio of total costs).

Strategy	S1	S2	S3	S4
Speedup	1.014 ± 0.013	1.013 ± 0.014	1.014 ± 0.013	1.013 ± 0.012

Appendix A.3. Backward Compatibility Details

Table A.9 provides the detailed backward compatibility verification across five diverse instances, confirming that the ZeroShot policy extraction introduces negligible error (average 1.32%, max 1.95%).

Table A.9: Backward compatibility: ZeroShot(C=1) vs original model.

Instance Diff(%)	Bays	Containers	Original Cost	ZeroShot Cost
R011606_001 1.95	1	70	4,807.2	4,713.6
R011606_003 1.63	1	70	4,872.0	4,793.0
R021606_001 1.01	2	140	13,067.0	12,935.4
R041606_001 0.61	4	280	31,252.0	31,061.0
R061606_001 1.39	6	430	49,421.0	48,736.0
Average 1.32	—	—	—	—

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